

**LAB Manual – 3**

Code/Course: 23CSE312/Distributed Systems

**Topic:** Video Streaming Service**[CO05] [BTL-3]****Student Name:** GARLAPATI TEJASWINI**Roll No.:** AM.SC.U4CSE23219**Batch/Section:** CSE C**Date:**14-02-2026

## 1 Introduction

This laboratory experiment focuses on developing a distributed application that implements a video streaming service using Python. In this experiment, a video file stored at the server side (for example, test.mp4) is streamed to the client over a network. The client receives the video data and displays each video frame as soon as it is transmitted by the server. The implementation makes use of socket programming for data transmission and the OpenCV library for video reading and playback.

## 2 Aim

To design and implement a distributed video streaming application in Python that streams video data from a server to one or more clients and displays the video frames at the client side in real time.

## 3 Objectives

- To understand the use of socket programming for multimedia data transmission.
- To implement video capture and playback using the OpenCV library.
- To develop a server capable of streaming video content to clients.
- To support video streaming to multiple clients using either an iterative or concurrent server model.
- To analyze the differences between video streaming communication and basic message-based communication.

## 4 System Requirements

- Operating System: Linux / Windows
- Programming Language: Python 3
- Libraries:
  - socket

- OpenCV (cv2)
- threading or equivalent (for concurrent server, if implemented)
- Hardware: System with camera/video playback capability

## 5 Task

Develop a distributed application implementing a video streaming service using Python. A given video at the server side (for example, test.mp4) is streamed and played back at the client end. The client should display each video frame as soon as it is transmitted from the server.

The implementation requires the use of the OpenCV library for video opening and reading operations, and socket programming for transmitting video data between the server and the client.

## 6 Functional Requirements

The application should be capable of the following:

- a. Providing streaming service to multiple clients. The server may be implemented either as an iterative server or a concurrent server.
- b. Maintaining a repository of video files at the server side, from which the client can choose a video to stream.

## 7 Submission Requirements

The submission should contain:

1. Python scripts for both server and client.
2. A document containing answers to the tasks along with screenshots of the output from both the server and the client.

## Results

### Server:

```
C:\Users\garla\Desktop\Sem 6 Labsheets\Distributed Systems\labsheet3>python server.py
Server started...
Waiting for clients...
Connected to ('127.0.0.1', 64882)
Streaming finished for ('127.0.0.1', 64882)
Connected to ('127.0.0.1', 52737)
Streaming finished for ('127.0.0.1', 52737)
█
```

## Client:

